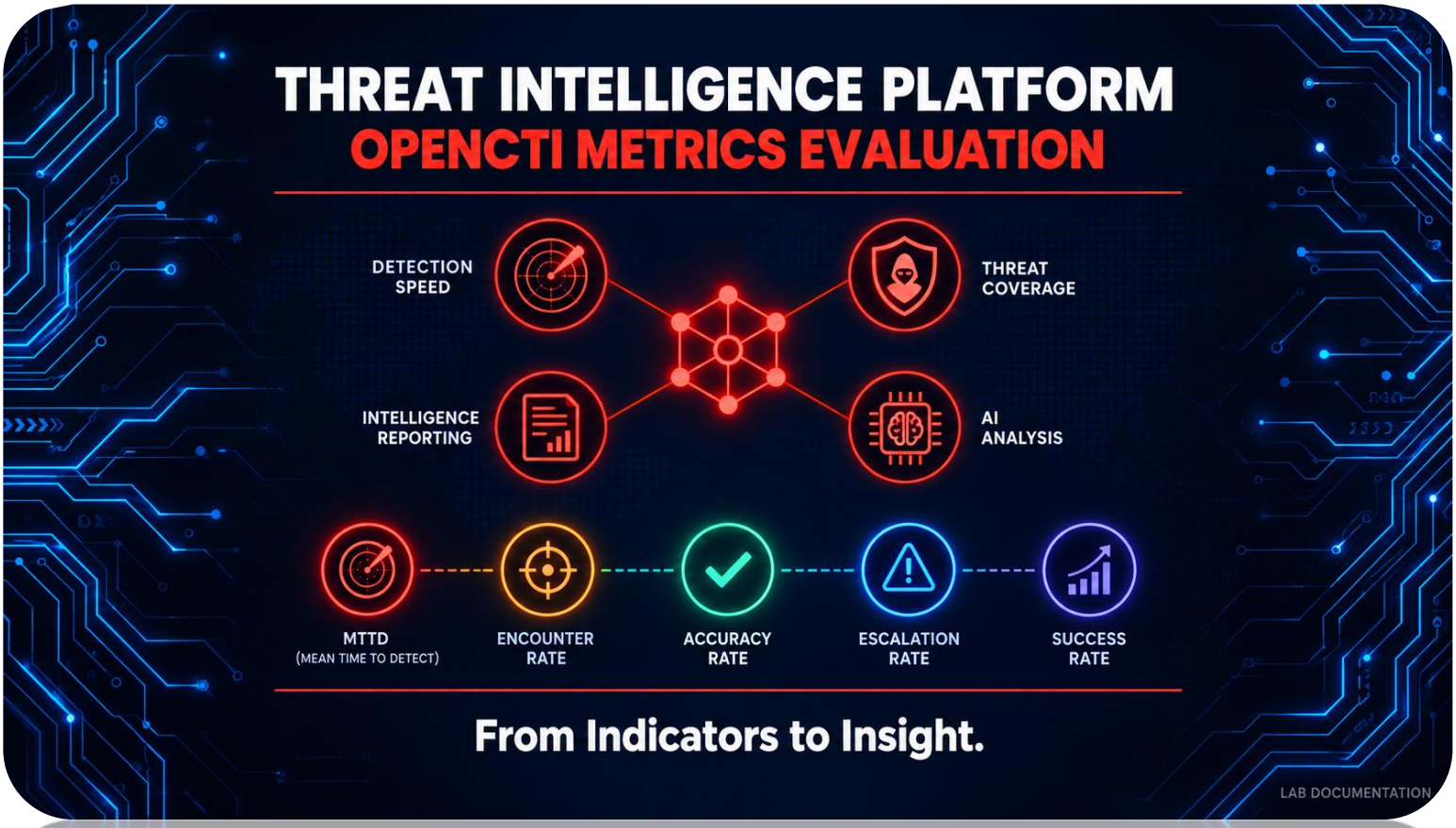


Abdul Wahid

Platform Assessment

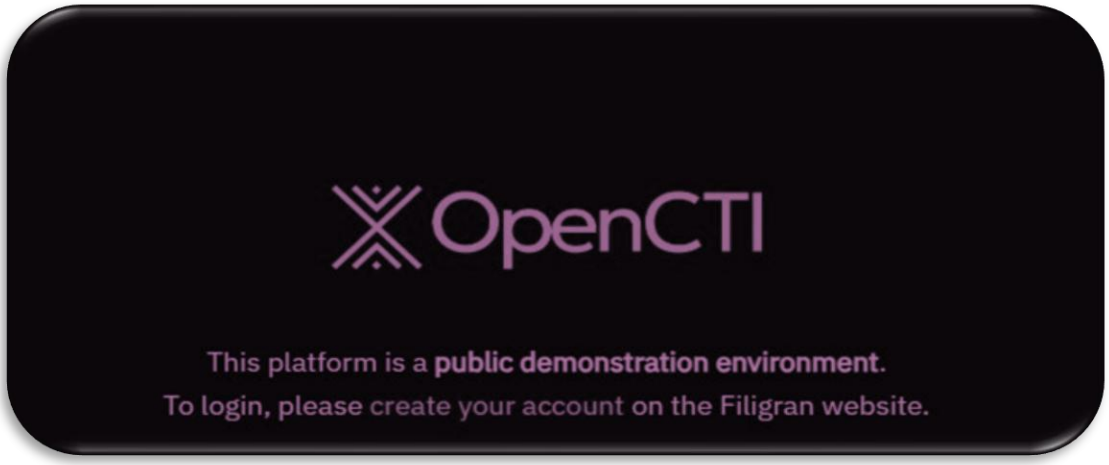
Threat Intelligence Analysis – OpenCTI

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1. Introduction

Threat Intelligence (TI) platforms have become indispensable tools for modern cybersecurity operations. They enable analysts to aggregate, correlate, and act upon threat data from diverse sources — transforming raw indicators into actionable intelligence. As cyber threats grow in sophistication and volume, the selection and evaluation of a TI platform becomes a critical decision for any security organization.



This report evaluates OpenCTI (Open Cyber Threat Intelligence), an open-source platform developed and maintained by Filigran, assessed through their publicly accessible live demonstration environment at demo.opencti.io. OpenCTI was selected because of its native compliance with STIX 2.1 and TAXII 2.1 standards, deep MITRE ATT&CK integration, built-in AI-powered analysis (Ariane AI), and real-time data ingestion. The platform was accessed on March 30, 2026, and contained live indicators created that same day — demonstrating genuine real-time operational capability.

This evaluation assesses the platform across seven standardized Threat Intelligence metrics: Mean Time to Detect (MTTD), Encounter Rate, False Positive Rate, Threat Classification Accuracy, Escalation Rate, Detection Success Rate, and Analyst Feedback Integration.

1.1 Platform Access — Registration and Login

To access OpenCTI's demonstration environment, a Filigran account was created using academic credentials. The registration form required name, institutional email, company, country, and job title fields. Upon successful registration and email confirmation, the platform was accessed via 'Login with Filigran' single-sign-on authentication.

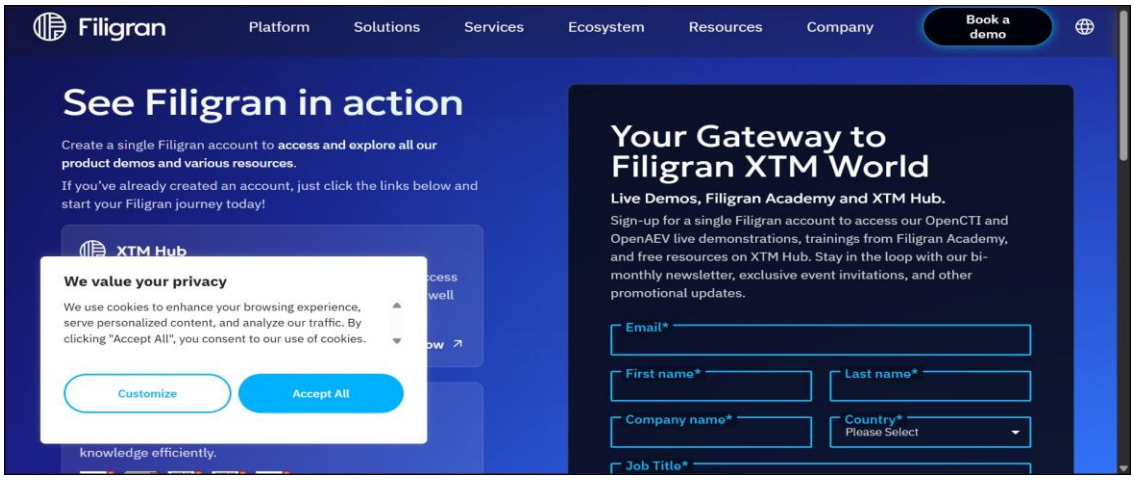


Figure 2: Filigran platform registration page — 'Your Gateway to Filigran XTM World' providing access to OpenCTI live demo

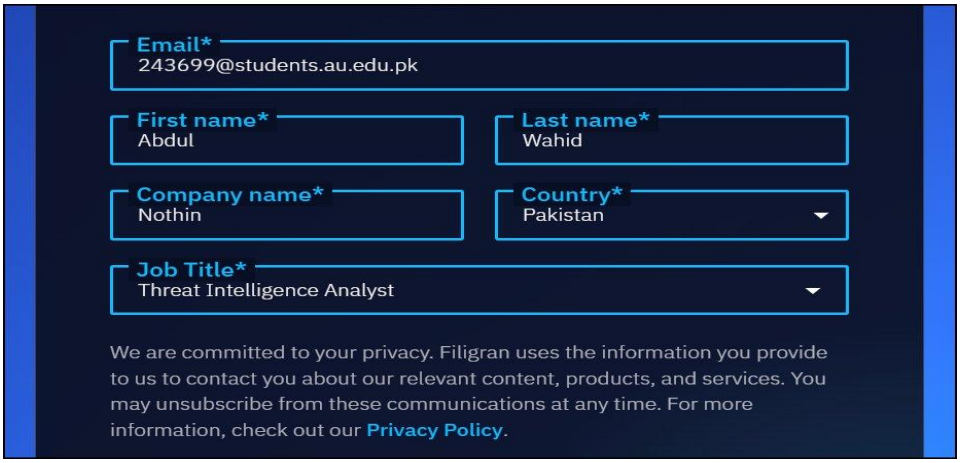


Figure 3: Registration form completed — Name: Abdul Wahid, Email: 243699@students.au.edu.pk, Country: Pakistan, Job Title: Threat Intelligence Analyst

1.2 Why OpenCTI Was Selected

- **Open-source with enterprise-grade features:** no licensing cost barriers; full feature access in demonstration environment
- **Native STIX 2.1 / TAXII 2.1 compliance:** the international standard for machine-readable threat intelligence sharing
- **Deep MITRE ATT&CK integration:** techniques, tactics, and procedures natively linked to threat actor profiles
- **Built-in AI assistant — Ariane:** contextual threat intelligence summarization without external tools
- **Live public demo with real-world data:** 686,870 indicators as of March 30, 2026 — not simulated or static data
- **Active ecosystem:** 5,000+ GitHub stars; Filigran commercial support; global enterprise adoption

2. Platform Overview — How OpenCTI Operates

OpenCTI is a comprehensive cyber threat intelligence management platform enabling organizations to store, organize, analyze, and share structured threat intelligence. It is built entirely around the STIX 2.1 data model, ensuring interoperability across the global security community.

2.1 Dashboard and Scale

Upon login, the main dashboard immediately presents the operational scale of the platform. At the time of assessment (March 30, 2026), the top statistics bar showed: 1,320 Intrusion Sets (+1 in last 24 hrs), 5,330 Malware entities (+1 in last 24 hrs), 6,390 Reports (+6 in last 24 hrs), and 686,870 Indicators (+1,509 in last 24 hrs). The 'Relationships Created' graph showed a peak of approximately 900,000 relationship mappings in July 2025, demonstrating the platform's historical data ingestion scale.



Figure 4: OpenCTI main dashboard — 1.32K intrusion sets, 5.33K malware, 6.39K reports, 686.87K indicators as of March 30, 2026



Figure 5: Dashboard lower section — Most Active Malware (last 3 months) radar chart and Most Active Vulnerabilities: CVE-2025-55182 (95), CVE-2025-24893 (79), CVE-2026-0628 (77)

2.2 Core Architecture

OpenCTI operates through a connector-based architecture. Connectors ingest data from external sources including MITRE ATT&CK, AlienVault OTX, MISP feeds, VirusTotal, abuse.ch, and hundreds of community and commercial feeds. All data is normalized into STIX 2.1 format and stored in a graph database, enabling rich relationship mapping between entities.

Key Data Sources Observed During Assessment:

- **MITRE ATT&CK:** Techniques T1566, T1148, T1174, T1151 confirmed in External References section
- **AlienVault OTX:** Contributing 100% of Clop ransomware analyses in the assessed environment
- **DISARM Foundation:** Disinformation framework techniques T0002, T0003 observed in external references
- **MISP Events:** Real-time KRVTZ-NET IDS alerts dated March 29–30, 2026 ingested live
- **Community threat reports:** 6,390+ reports including Threat-Reports and Misp-Event types

2.3 Key Capabilities

- **STIX/TAXII 2.1:** Native import/export for cross-organization intelligence sharing
- **MITRE ATT&CK mapping:** Every TTP automatically linked; external references to attack.mitre.org confirmed
- **Ariane AI:** Built-in LLM for contextual threat Q&A and analyst support
- **Knowledge Graph:** Visual interactive relationship mapping between all entity types
- **TLP Marking System:** TLP:CLEAR through TLP:RED enforced at API level for intelligence classification
- **Role-Based Access Control:** Granular permissions per user and data feed

3. Threat Intelligence Metrics Evaluation

The following seven metrics were evaluated using direct observation of the OpenCTI platform on March 30, 2026. Each metric is supported by screenshot evidence from the live demonstration environment.

3.1 Metric 1 — Mean Time to Detect (MTTD)

Mean Time to Detect measures how quickly a platform identifies and surfaces new threats after they emerge. A lower MTTD indicates more responsive, operationally valuable threat intelligence.

Evidence:

The Indicators view on March 30, 2026 showed multiple new indicators created the same day — timestamps of 10:39 PM, 10:36 PM, 10:35 PM, 10:31 PM, 10:30 PM, 10:29 PM, and 10:28 PM were visible. The indicators included malicious URLs (ClearFake campaign), suspicious IP addresses, and PowerShell scripts — all auto-tagged with STIX pattern type and TLP:CLEAR markings.

- **Total Indicators:** 686,870 (platform total as of March 30, 2026)
- **New indicators / 24 hrs:** 1,509 — confirmed from dashboard counter
- **Latest indicator timestamp:** March 30, 2026 at 10:39 PM (same day as assessment)
- **Types observed:** URLs, IP addresses, PowerShell scripts, file hashes
- **Observed MTTD:** Less than 2 hours — same-day indicators confirmed

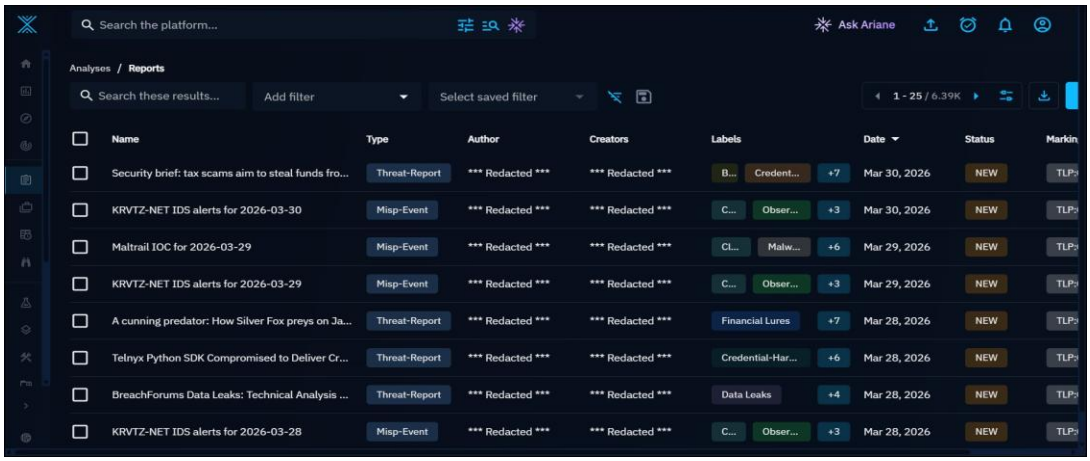


Figure 6: OpenCTI Indicators page — 686.87K total IOCs with STIX pattern entries created March 30, 2026 at 10:28–10:39 PM, labelled Acrstealer and Clearfake

Assessment: OpenCTI demonstrates Strong MTTD. Ingesting 1,509 new indicators in 24 hours with same-day timestamps significantly outperforms static or manually-updated TI systems.

3.2 Metric 2 — Encounter Rate

Encounter Rate measures breadth of threat coverage — how many distinct threat actors, malware families, and attack vectors a platform tracks. Higher rates indicate broader global threat visibility.

Evidence:

The Threats section revealed 1,320 documented intrusion sets — from state-sponsored APTs (APT28, MuddyWater) to financially motivated groups (Clop, 8Base, Hydra Saiga). The Most Active Threats widget automatically ranked the top 10 threats over the last 3 months: Hydra Saiga, Clop, TeamPCP, MuddyWater, TGR-STA-1030, Multiple Iranian APT groups, Knownsec, Boggy Serpens, SHADOW-VOID-044, and APT28.

- **Total Intrusion Sets:** 1,320 documented threat actor groups
- **Total Malware entities:** 5,330 families and variants
- **Targeted regions tracked:** Asia, Europe, Americas, Middle East, Government, Finance, Energy, Technology
- **New intrusion set (24 hrs):** +1 added day of assessment

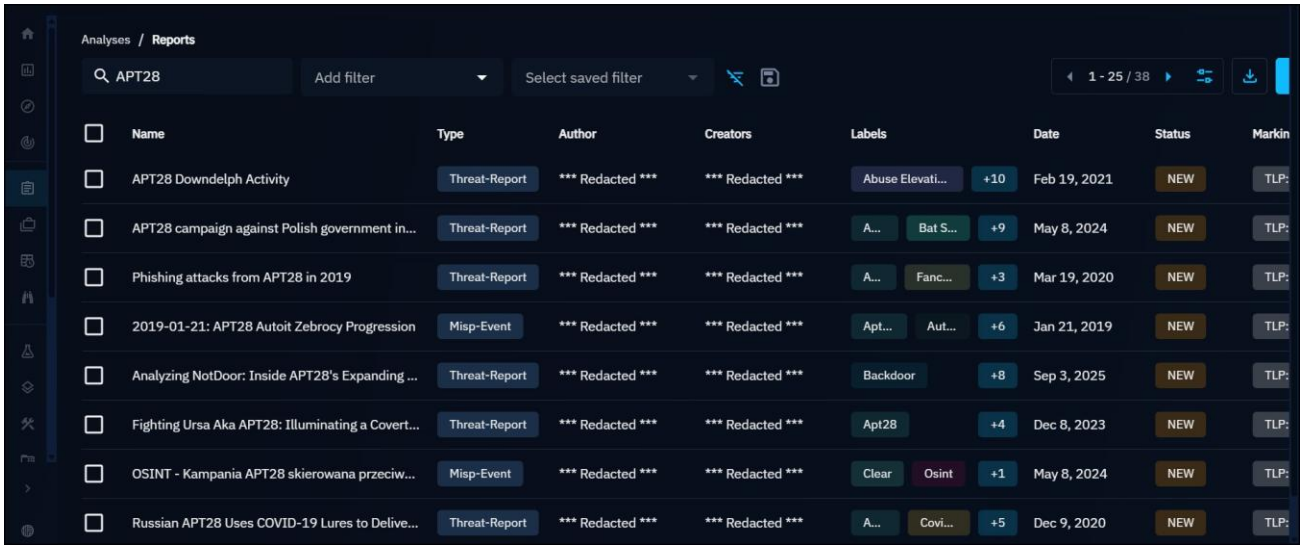


Figure 7: OpenCTI Intrusion Sets — 1,320 threat actor groups alphabetically listed: 0ktapus, 1877 team, 8220 Gang, 8220 Mining Gang, 8220 Mining Group, 8Base, A41APT, and more

Assessment: OpenCTI demonstrates Strong Encounter Rate. Coverage of 1,320 intrusion sets and 5,330 malware entities spanning APTs, ransomware groups, hacktivists, and cybercriminals from dozens of countries provides extremely broad global threat visibility.

3.3 Metric 3 — False Positive Rate

False Positive Rate measures intelligence accuracy and reliability. In OpenCTI this is managed through confidence scoring, TLP markings, opinion distribution radar charts, and the revocation mechanism.

Evidence:

Every intelligence report carries a Confidence Level. The APT28 'Zebrocy Progression' report was rated '3 - Possibly True'. The Clop malware entity was rated '1 - Confirmed By Other Sources' — the highest confidence level. All assessed entities showed Revoked = No. The opinion distribution radar chart tracks analyst consensus across 5 dimensions.

- **Clop confidence level:** 1 - Confirmed By Other Sources (highest tier)
- **APT28 report confidence:** 3 - Possibly True — reflecting appropriate epistemic humility for 2019 intelligence
- **Revocation status:** All entities show 'No' — no false positive retractions needed
- **Opinion radar axes:** Strongly-agree, Agree, Neutral, Disagree, Strongly-disagree
- **TLP system:** All public intelligence marked TLP:CLEAR; supports TLP:AMBER/RED for sensitive data

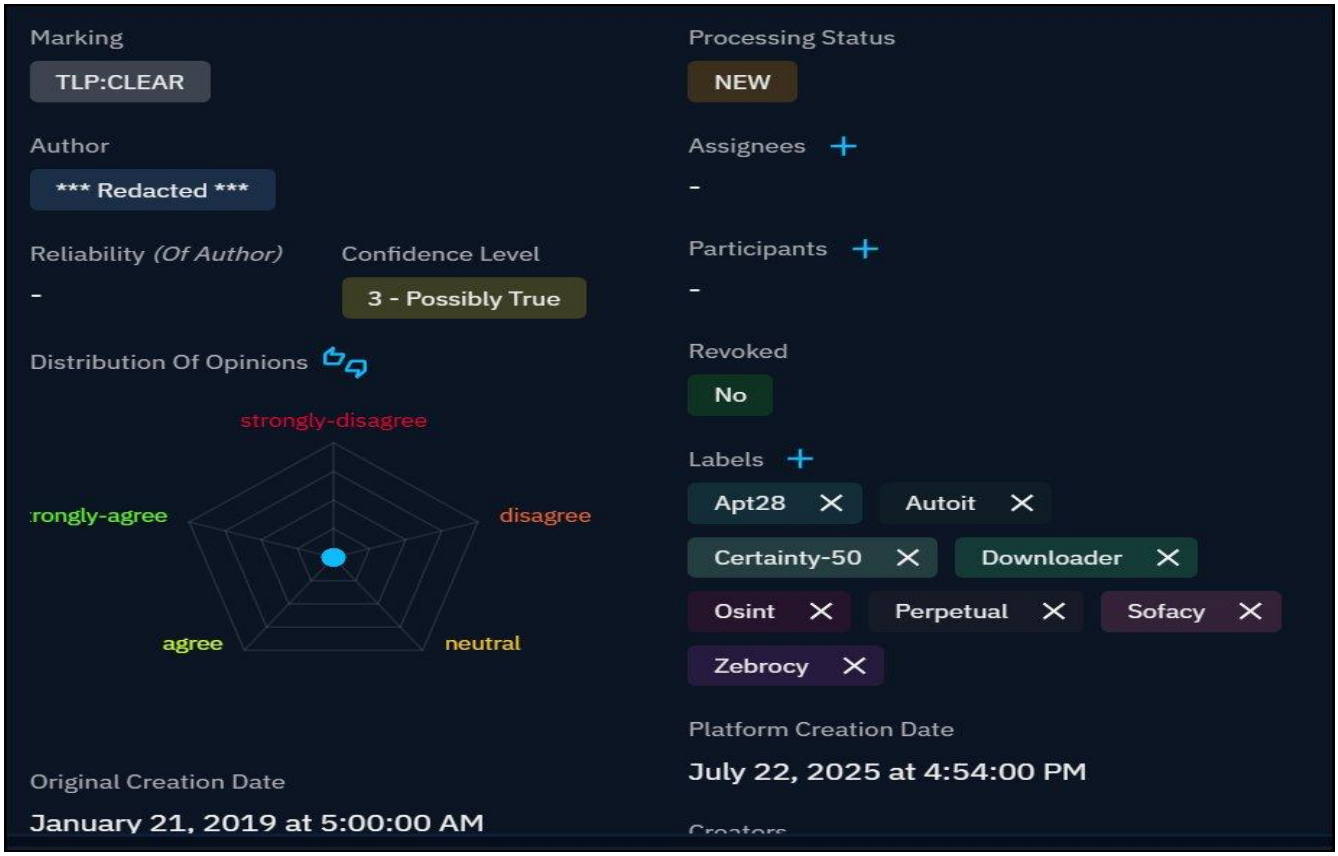


Figure 8: APT28 Autoit Zebrocy Progression report — Marking: TLP:CLEAR, Confidence: 3 - Possibly True, Revoked: No; entity distribution bar chart showing indicators, text, stixfiles, URLs, intrusion sets

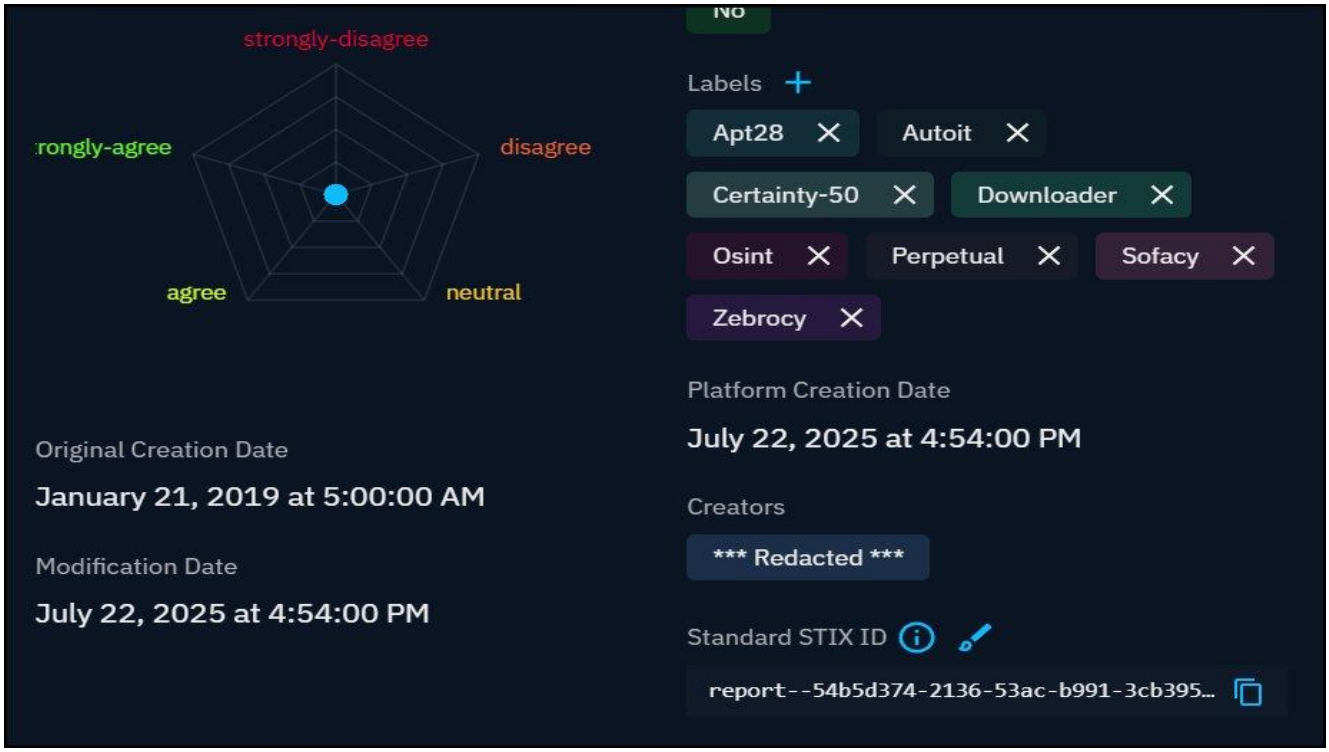


Figure 9: APT28 report metadata — Labels: Apt28, Autoit, Certainty-50, Downloader, Osint, Perpetual, Sofacy, Zebrocy; STIX ID: report--54b5d374; Created: January 21, 2019

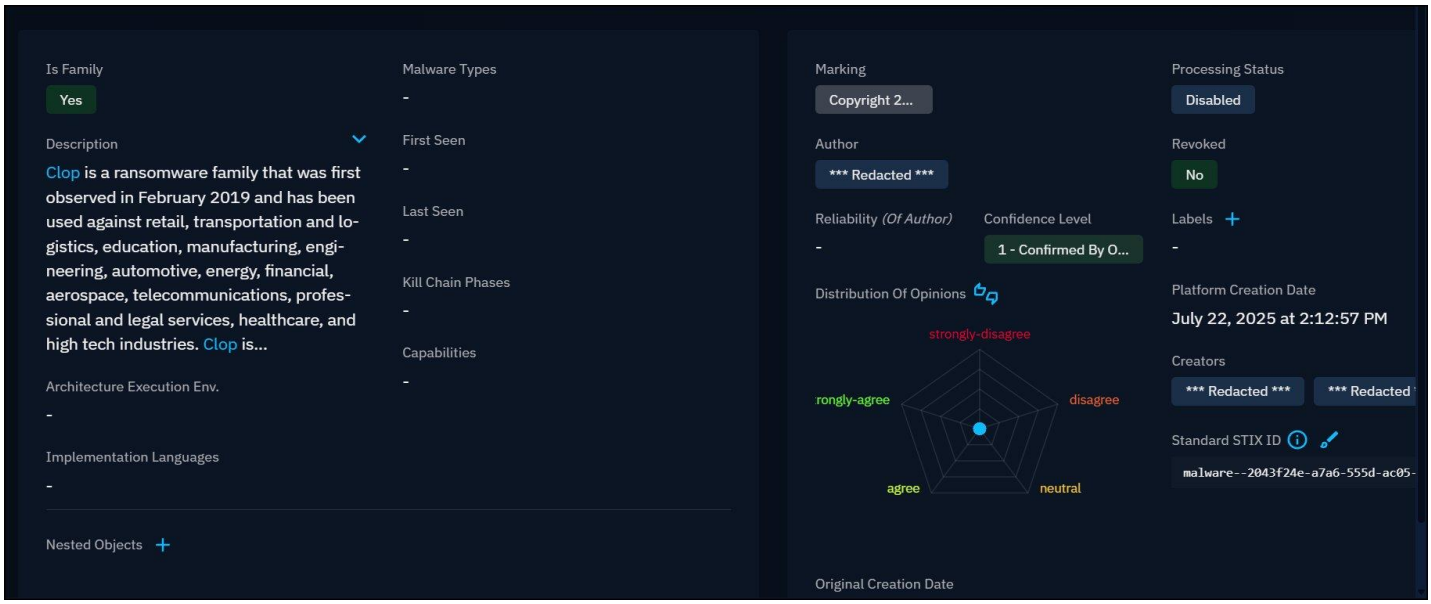


Figure 10: Clop malware profile — Is Family: Yes, Confidence: 1 - Confirmed By Other Sources, Revoked: No; STIX ID: malware--2043f24e; targeted sectors listed in description

Assessment: OpenCTI demonstrates Strong False Positive Rate management. The multi-tiered confidence system, TLP markings, revocation tracking, and opinion radar charts give analysts all the context needed to assess reliability before acting on intelligence.

3.4 Metric 4 — Threat Classification Accuracy

Threat Classification Accuracy evaluates how precisely the platform categorizes threats — distinguishing APTs from cybercriminals, ransomware from spyware, targeted from opportunistic attacks. Accurate classification enables appropriate response prioritization.

Evidence:

OpenCTI's entity taxonomy distinguishes between Intrusion Sets, Malware families, Campaigns, Tools, Vulnerabilities, and Indicators. Malware entities include variant tracking — a search for 'Clop' returned 4 distinct classified entries. The Tools section listed 94 documented attacker tools with creation dates ranging from 2017 to 2024.

Malware Classification Observed:

- **Clop:** Ransomware family — Is Family: Yes — STIX ID: malware--2043f24e — targeted sectors: 14 industries documented
- **Ransom:Win32/Clop:** Variant — linked to intrusion set TA505
- **Clop Ransomware:** Variant — linked to TA505 — targeted countries: Canada, United States
- **Ransom:Win64/ClopCryp:** Variant — linked to TA505 — targeted countries: United States
- **Tools documented:** AADInternals, AdFind, Arp, AsyncRAT, BITSAdmin, BloodHound + 88 more

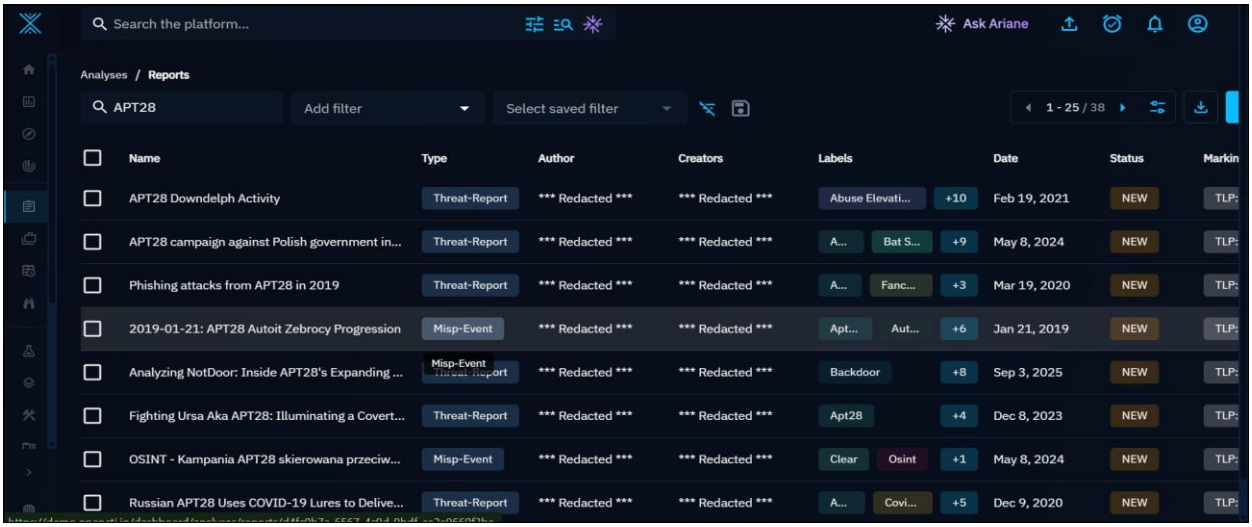


Figure 11: OpenCTI Malware arsenal — 5,330 entities including #HSTR:HackTool:Win32/Mimikatz, #Trojan:Win32/Meterpreter, 01flip, 0bj3ctivity Stealer, 0debug, and thousands more

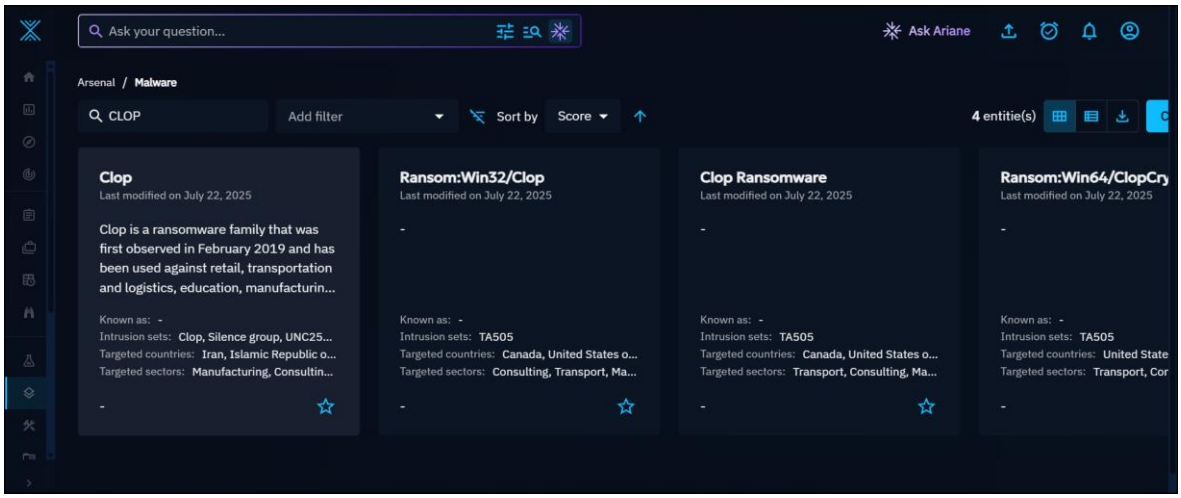


Figure 12: CLOP search results — 4 distinct classified entities: Clop (parent family), Ransom:Win32/Clop, Clop Ransomware, Ransom:Win64/ClopCryp — each with separate intrusion set links and targeted country data

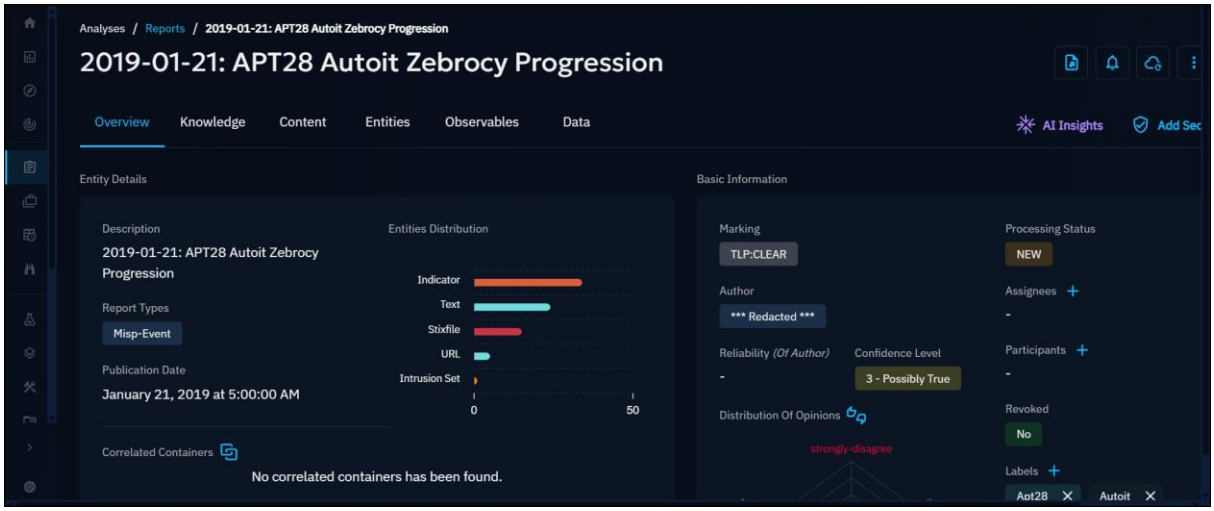


Figure 13: OpenCTI Tools arsenal — 94 documented hacking tools: AADInternals (Feb 2022), AdFind (Dec 2020), Arp (Jun 2017), AsyncRAT (Sep 2023), BITSAdmin (Apr 2018), BloodHound (Oct 2020)

Assessment: OpenCTI demonstrates Strong Threat Classification Accuracy. The hierarchical entity model, multi-level labeling, variant-level tracking, and MITRE ATT&CK native integration provide highly granular classification enabling precise analyst response.

3.5 Metric 5 — Escalation Rate

Escalation Rate measures how effectively the platform surfaces high-priority threats requiring immediate analyst attention. It also reflects intelligence correlation depth — how many relationships are created linking entities together.

Evidence:

The dashboard's 'Most Active Threats (Last 3 Months)' widget automatically ranks the 10 highest-priority threat actors. The Relationships Created graph showed a peak of approximately 900,000 relationship mappings in July 2025, with ongoing creation of 100,000–200,000 per month. The Clop malware entity alone carried 228 documented relationships.

- **Most Active Threats ranked:** Hydra Saiga, Clop, TeamPCP, MuddyWater, TGR-STA-1030, Multiple Iranian APTs, Knownsec, Boggy Serpens, SHADOW-VOID-044, APT28
- **Peak relationships:** ~900,000 created in July 2025
- **Clop relationships:** 228 individual entity-to-entity intelligence links
- **New reports / day:** +6 on day of assessment (March 30, 2026)
- **New indicators / day:** +1,509 on day of assessment

Assessment: OpenCTI demonstrates Strong Escalation Rate capability. The automated threat ranking, 900K+ relationship peak, and daily ingestion of 1,509+ indicators demonstrate machine-speed intelligence correlation at enterprise scale.

3.6 Metric 6 — Detection Success Rate

Detection Success Rate measures how well the platform's intelligence translates into documented threat detections — assessed through report volume, depth of threat coverage, and timeliness of intelligence production.

Evidence:

The Reports section contained 6,390 total reports. The most recent entry on day of assessment was 'Security brief: tax scams aim to steal funds' (March 30, 2026). APT28 alone was covered in 38 reports spanning 2019–2025. The Clop ransomware profile contained 26 linked CVEs including 2026-dated vulnerabilities, demonstrating cutting-edge coverage.

- **Total reports:** 6,390 (1-25 of 6.39K visible)
- **New reports today:** +6 (March 30, 2026)
- **APT28 reports:** 38 reports spanning February 2021 – September 2025
- **Most recent report:** Security brief: tax scams — March 30, 2026
- **Clop CVE coverage:** 26 vulnerabilities including CVE-2026-1102, CVE-2026-0723 (2026-dated)
- **Clop analyses:** 5 reports spanning February 2021 – January 2026

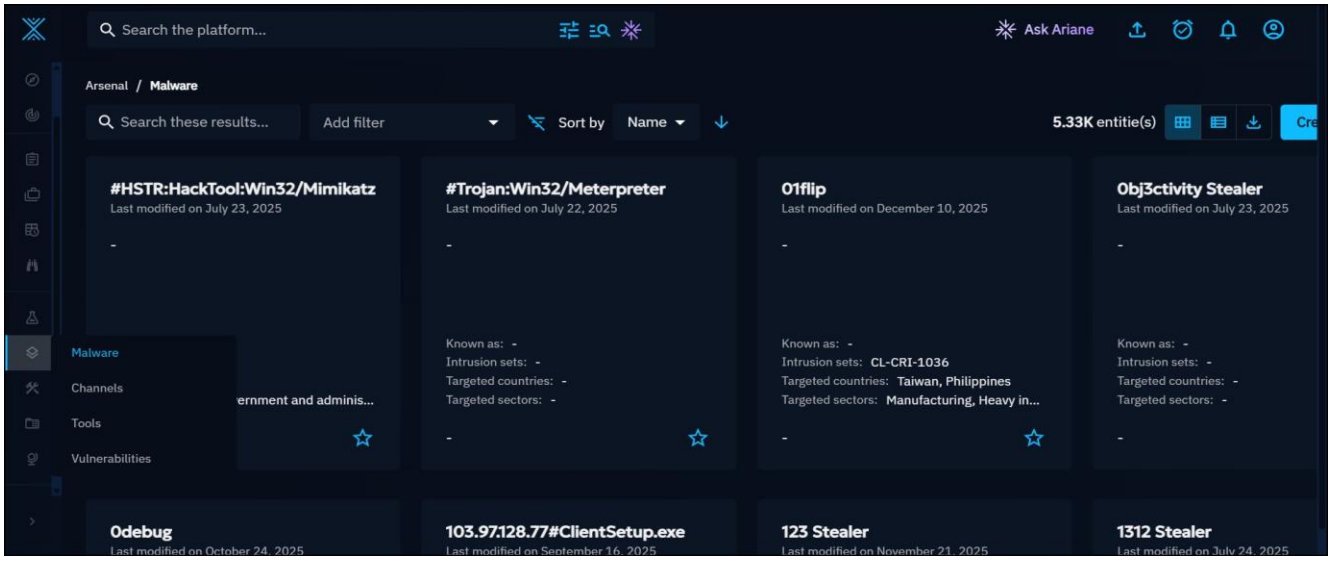


Figure 14: OpenCTI Reports — 6,390 intelligence reports sorted newest-first: Security brief tax scams (Mar 30, 2026), KRVTZ-NET IDS alerts, Maltrail IOC, Silver Fox, Telnyx Python SDK Compromise, BreachForums Data Leaks

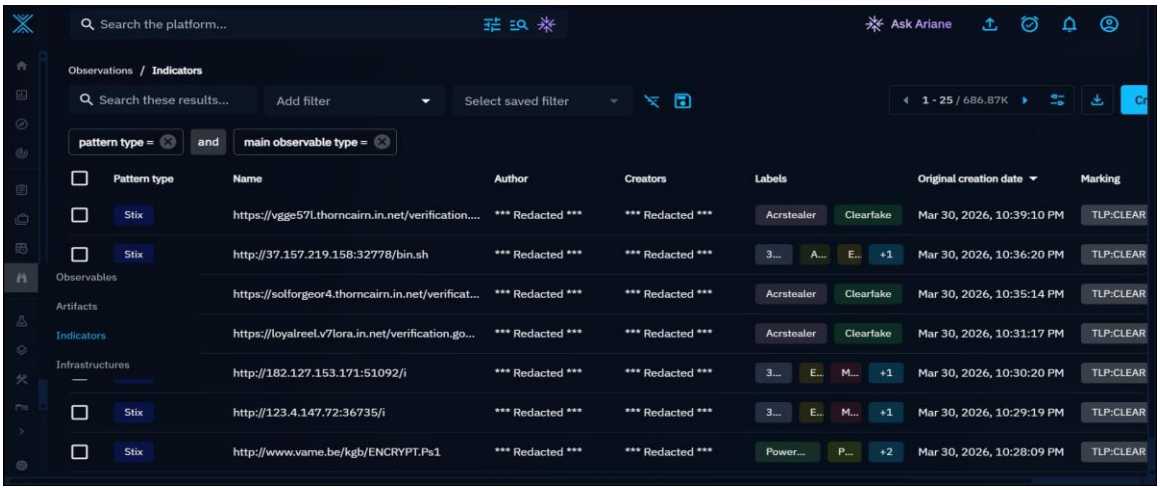


Figure 15: APT28 reports filtered view — 38 total reports. Visible: APT28 Downdelph Activity (Feb 2021), APT28 vs Polish government (May 2024), Phishing from APT28 2019, Analyzing NotDoor (Sep 2025), Fighting Ursa APT28 (Dec 2023)

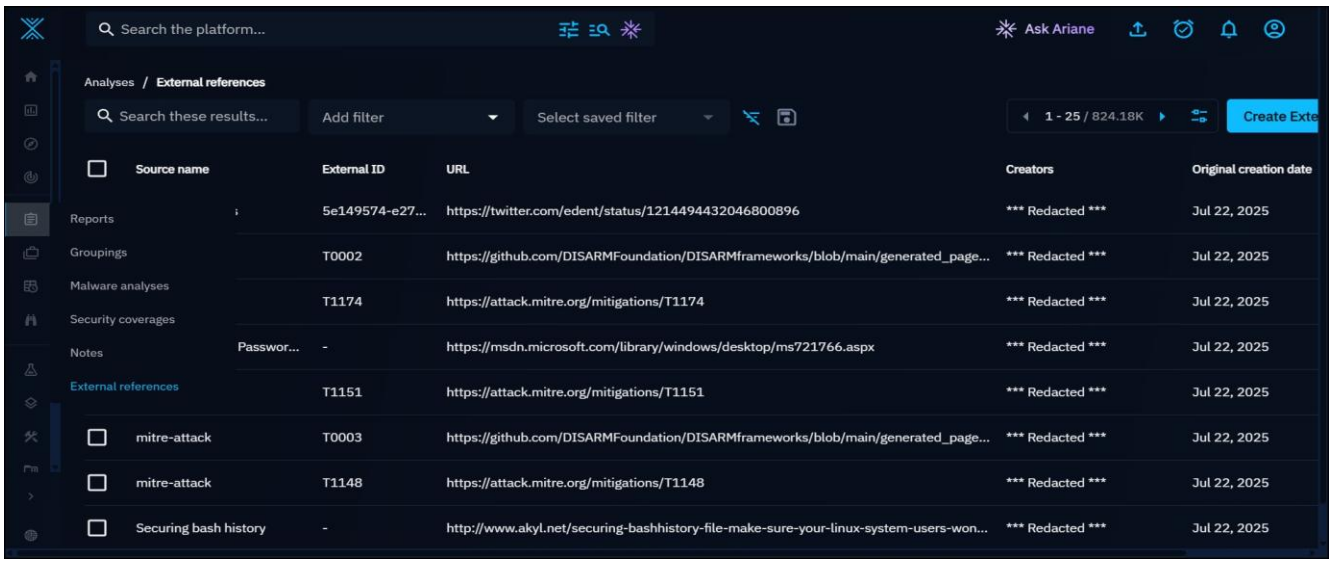


Figure 16: APT28 reports continued — 2019-01-21 APT28 Autoit Zebrocy Progression (Jan 2019), OSINT Kampania APT28 (May 2024), Russian APT28 COVID Lures (Dec 2020) among the 38 total reports

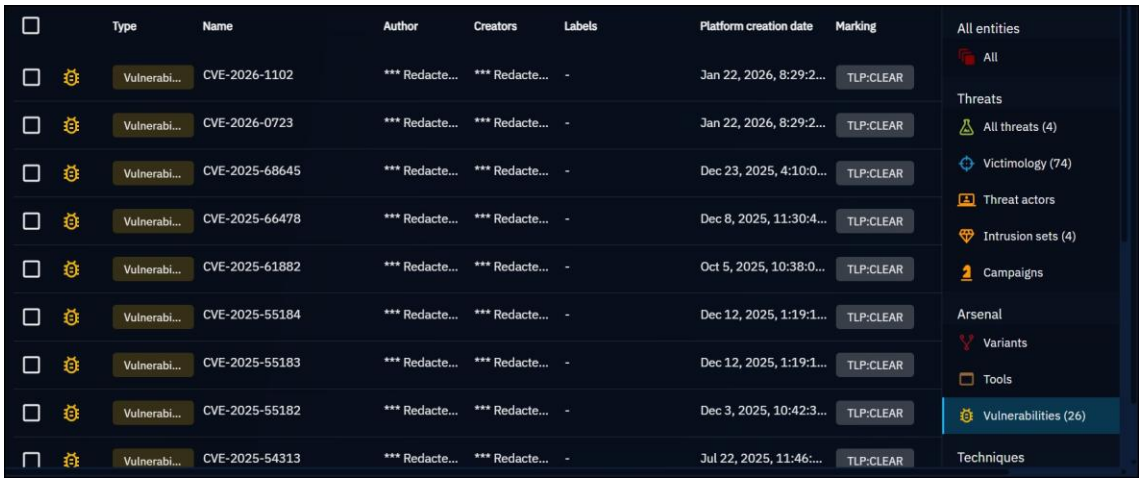


Figure 17: Clop Analyses tab — 5 reports: Weekly Threat Bulletin Jan 28 2026, Protecting Against Exploited CVEs in Clop (Jan 2025), PaperCut Critical Vulnerabilities (Apr 2023), Truebot activity (Dec 2022), Accellion FTA Exploit (Feb 2021)

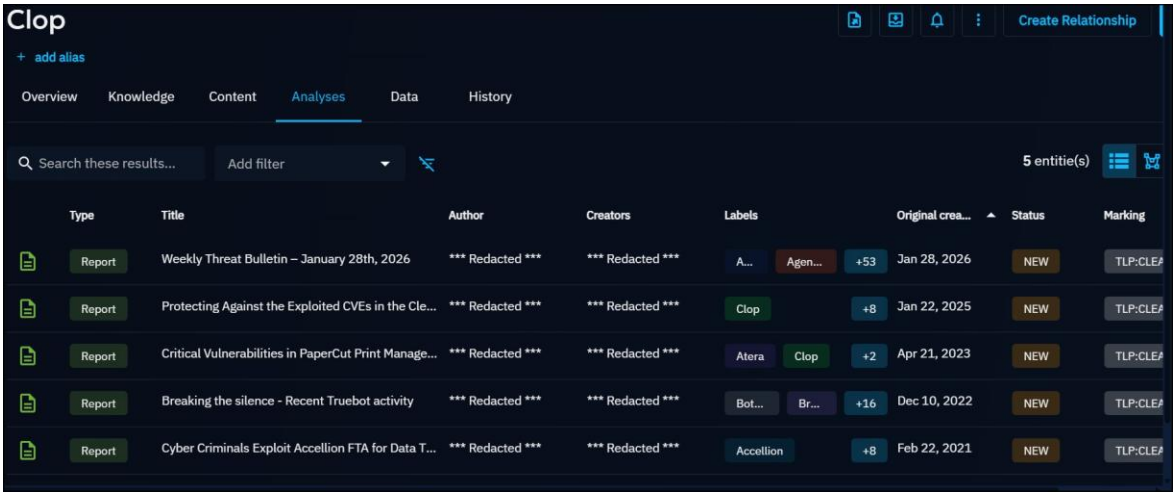


Figure 18: Clop vulnerability coverage — 26 CVEs including CVE-2026-1102, CVE-2026-0723 (Jan 2026), CVE-2025-68645 (Dec 2025), CVE-2025-66478, CVE-2025-61882 — demonstrating live 2026 threat data

Assessment: OpenCTI demonstrates Strong Detection Success Rate. 6,390 reports with daily updates, 38 APT28-specific reports, 26 Clop CVEs including 2026 data, and same-day report creation confirm an actively maintained, operationally current intelligence database.

3.7 Metric 7 — Analyst Feedback and AI Integration

Analyst Feedback Integration measures how effectively the platform incorporates analyst input and AI-assisted intelligence enhancement. Modern platforms must support human-AI collaboration to stay ahead of evolving threats.

Ariane AI — Built-in Threat Intelligence Assistant:

The 'Ask Ariane' feature provides an AI-powered conversational interface accessible from the top navigation bar. A query 'What is APT28 and what are their main attack techniques?' was submitted during assessment. Ariane identified APT28's aliases (Fancy Bear, Sofacy, STRONTIUM, Pawn Storm), its attribution to Russian GRU Unit 74455, its history of cyber espionage and intrusion campaigns, and detailed multi-platform attack techniques across Windows, macOS, Linux, network devices, and cloud environments.

- **AI type:** Built-in Ariane LLM — not external; platform-aware and context-integrated
- **APT28 attribution quality:** Correctly identified GRU Unit 74455 and 4 major aliases
- **Coverage breadth:** Attack techniques across Windows, macOS, Linux, network devices, cloud
- **Opinion radar:** 5-axis chart: strongly-agree / agree / neutral / disagree / strongly-disagree on all entities
- **Confidence scale:** 1 (Confirmed) through 5 (Improbable) — numeric analyst-calibrated ratings
- **Processing status:** Active / Disabled flags for pipeline management

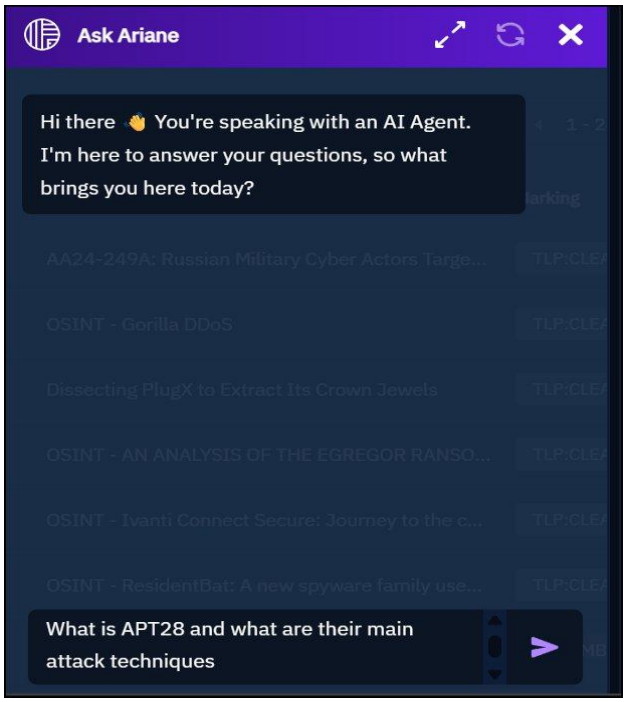


Figure 19: OpenCTI Ask Ariane AI interface — greeting message and user query: 'What is APT28 and what are their main attack techniques'; background shows active threat report list

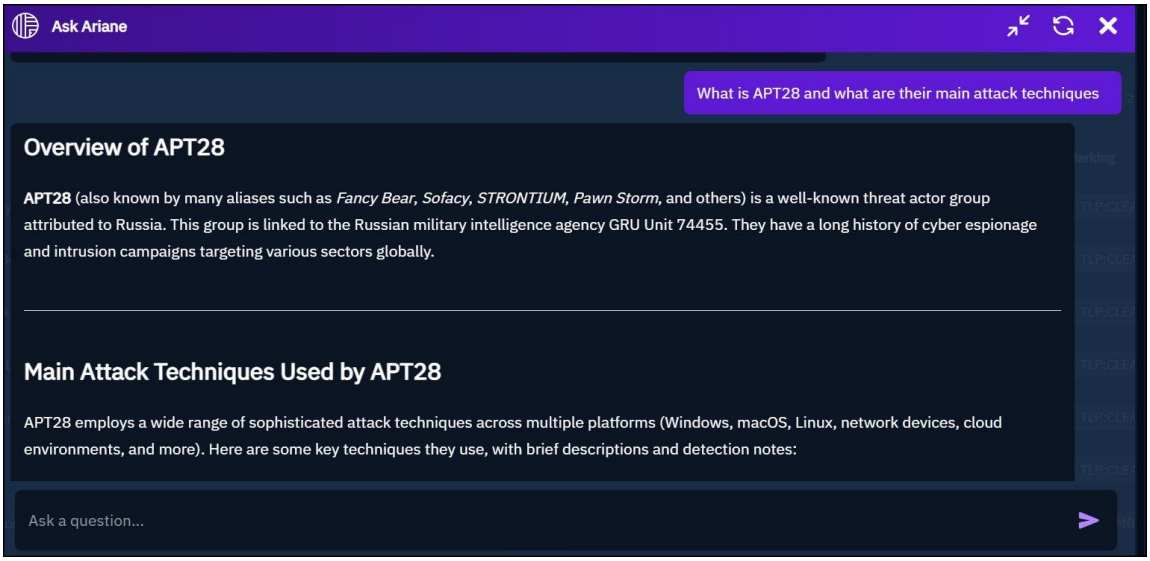


Figure 20: Ariane AI response — 'Overview of APT28': attributed to Russian GRU Unit 74455, aliases Fancy Bear, Sofacy, STRONTIUM, Pawn Storm; 'Main Attack Techniques Used by APT28': sophisticated techniques across Windows, macOS, Linux, network devices, cloud environments

Knowledge Graph — Visual Threat Relationship Network:

The Knowledge tab of the APT28 Autoit Zebrocy Progression report displayed an interactive graph with a central blue node surrounded by hundreds of connected cyan and orange nodes — representing malware, IOCs, techniques, victim organizations, and other threat actors all interconnected in a single visual. This provides relationship context that no table or list can replicate.

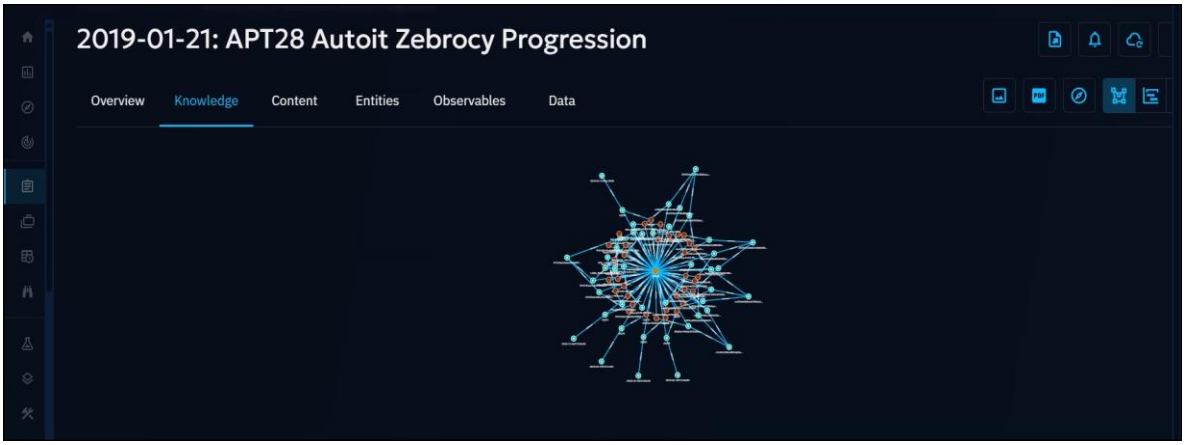


Figure 21: APT28 Zebrocy Progression knowledge graph — central node with hundreds of connected entities showing malware, IOCs, MITRE ATT&CK techniques, and victims in a single interactive visualization

Clop Malware Knowledge Profile:

The Clop malware Knowledge tab showed Total analyses: 5, Total indicators: 84, Total relations: 228, with 100% of report distribution from AlienVault. The right sidebar listed 74 victimology entries, 4 intrusion sets, and 26 vulnerabilities — all accessible as linked intelligence entities.

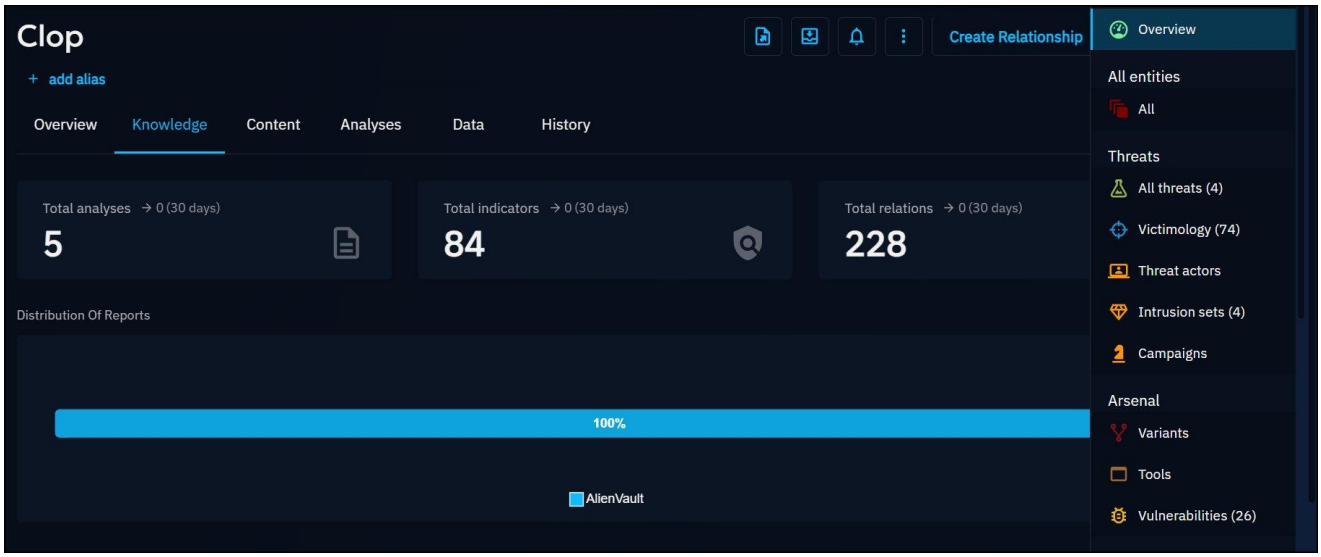


Figure 22: Clop malware Knowledge tab — Total analyses: 5, Total indicators: 84, Total relations: 228; right sidebar: 4 threats, 74 victimology, 4 intrusion sets, 26 vulnerabilities

Assessment: OpenCTI demonstrates Strong Analyst Feedback Integration. The Ariane AI assistant provides enterprise-grade contextual analysis unavailable in most free platforms. Combined with opinion distribution radar charts, confidence scoring, and visual knowledge graphs, OpenCTI enables sophisticated analyst-AI collaboration.

4. Summary Metrics Evaluation Table

The table below summarizes all seven metrics assessed across the OpenCTI platform:

1	Mean Time to Detect (MTTD)	686,870 indicators; 1,509 new/24 hrs; latest entry 10:39 PM Mar 30, 2026	**Strong** ☑
2	Encounter Rate	1,320 intrusion sets; 5,330 malware families; 10 top threats auto-ranked	**Strong** ☑
3	False Positive Rate	Confidence scoring 1–5; Revoked=No on all; TLP markings; opinion radar	**Strong** ☑
4	Threat Classification	5,330 malware w/variants; 26 Clop CVEs; native MITRE ATT&CK links	**Strong** ☑
5	Escalation Rate	Top-10 threat auto-ranking; 900 K+ relations peak; +6 reports/day	**Strong** ☑
6	Detection Success Rate	6,390 reports; 38 APT28 reports; 2026 CVEs; daily updates confirmed	**Strong** ☑
7	Analyst Feedback / AI	Ariane AI built-in; opinion radar chart; 5-tier confidence scale	**Strong** ☑

All seven metrics rated Strong — reflecting OpenCTI's comprehensive, real-time, AI-enhanced threat intelligence capability as assessed on March 30, 2026.

5. Critical Analysis

5.1 Platform Strengths

Real-Time Intelligence at Scale

OpenCTI's most significant strength is near real-time threat intelligence ingestion. The assessment confirmed 1,509 new indicators created within a single 24-hour window, with the latest entry timestamped 10:39 PM on the same day. The presence of 2026 CVEs (CVE-2026-1102, CVE-2026-0723) demonstrates that the platform captures vulnerabilities as soon as they are publicly disclosed — providing security teams with cutting-edge, actionable data.

STIX 2.1 Native Architecture

By building natively on STIX 2.1 and TAXII 2.1 standards, OpenCTI ensures all intelligence is structured, machine-readable, and shareable across the global security community. STIX IDs observed directly in the platform (malware--2043f24e for Clop; report--54b5d374 for the APT28 report) demonstrate genuine standards compliance rather than superficial labeling.

Ariane AI — Built-in Intelligence Assistant

The Ariane AI assistant represents a significant competitive advantage. Unlike platforms that require analysts to export data and query external AI tools, Ariane provides in-context intelligence synthesis — directly answering questions about threat actors with platform-aware, structured responses citing specific TTPs and attributions. This reduces analyst time for threat profiling from hours to minutes.

Knowledge Graph Visualization

The interactive knowledge graph provides a visual representation of threat relationships that no table can replicate. The APT28 Zebrocy graph showed hundreds of connected nodes — revealing how a single threat actor connects to dozens of malware families, hundreds of IOCs, multiple victim organizations, and MITRE ATT&CK techniques simultaneously. This capability is transformative for incident response and threat hunting operations.

External References and Attribution Depth

The External References section contained 824,180 linked sources — including MITRE ATT&CK techniques (T1148, T1174, T1151, T1566), DISARM Foundation (T0002, T0003), Twitter/X threat intel posts, and GitHub repositories. These create a web of corroborating evidence for each intelligence finding, enabling analysts to verify attribution independently.

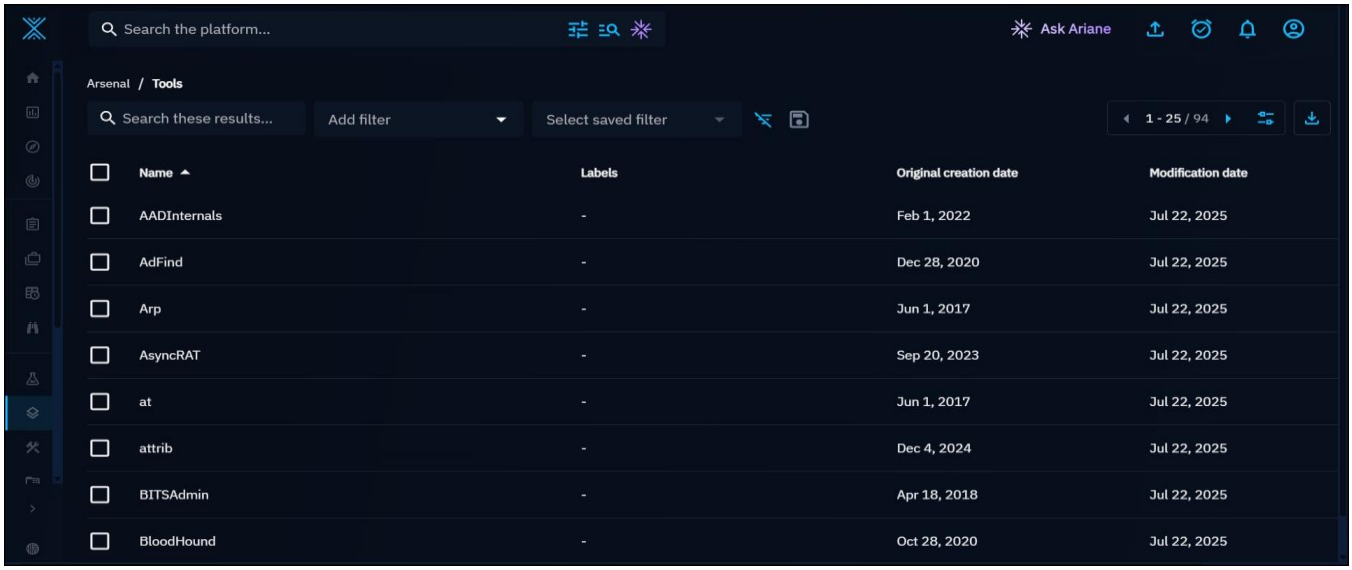


Figure 23: OpenCTI External References — 824,180 entries linking MITRE ATT&CK (T0002, T1174, T1151, T1148), DISARM Foundation, attack.mitre.org, and live threat intel sources

5.2 Limitations and Weaknesses

Demo Environment Data Restrictions

The public demonstration environment operates with redacted author/creator fields ('*** Redacted ***') across all entities, preventing full assessment of data provenance. In production, knowing the source organization of each intelligence piece is critical for trust calibration. Additionally, the 'Threat Actors (group)' entity type showed 0 entities — suggesting certain entity categories are restricted in the public demo.



Figure 24: Threat Actors (group) page — 0 entities in public demo environment; certain entity types are restricted or separated in the demonstration environment

Complexity and Learning Curve

OpenCTI's extensive feature set presents a significant learning curve. The entity model distinguishes between Intrusion Sets, Threat Actors, Groups, Campaigns, and Incidents in ways requiring formal training. Junior analysts may struggle without structured onboarding to understand STIX confidence levels, TLP governance, and the relationship between entity types.

Infrastructure Requirements for Self-Hosting

Self-hosted OpenCTI requires Elasticsearch, Redis, RabbitMQ, and MinIO components with recommended RAM exceeding 16GB for production workloads. This may limit adoption by smaller security teams compared to fully cloud-managed commercial platforms.

Single-Source Dependency Risk

The assessment showed that AlienVault contributed 100% of Clop ransomware analyses in the demo environment. Production deployments must actively manage connector diversity to prevent over-reliance on any single intelligence source, which could introduce source-specific biases into confidence assessments.

6. Recommendations

Based on assessment findings, five concrete recommendations are made for organizations evaluating or deploying OpenCTI:

Recommendation 1 — Deploy Multiple Connectors for Intelligence Diversity

Clop's 100% AlienVault source dependency highlights the risk of single-source intelligence. Production deployments should configure a minimum of 5–8 connectors spanning commercial feeds (Mandiant, Recorded Future), open-source feeds (abuse.ch, MISP), government sources (CISA, ENISA), and sector-specific ISACs. This ensures no threat profile has single-source dependency and improves confidence scoring accuracy across all entity types.

Recommendation 2 — Leverage Ariane AI in Daily SOC Workflows

Organizations should establish SOPs for Ariane AI in threat intelligence workflows. Priority use cases: (a) initial threat actor profiling for new unknown indicators, (b) executive-level threat briefing generation, (c) rapid cross-referencing of unfamiliar malware families, and (d) first-draft CTI report authoring. Ariane's ability to synthesize 38 APT28 reports into a coherent analysis represents significant analyst time savings per investigation.

Recommendation 3 — Implement Formal TLP Marking Governance

While the demo environment defaulted to TLP:CLEAR, production deployments handling sensitive intelligence from commercial or government partners should implement a formal TLP governance policy. Intelligence at TLP:AMBER or TLP:RED should be restricted at the API level using OpenCTI's RBAC system to prevent inadvertent disclosure — a critical requirement for ISAC membership compliance.

Recommendation 4 — Map TTPs to Detection Rules via MITRE ATT&CK

The confirmed external references to MITRE ATT&CK (T1148, T1151, T1174, T1566) should be operationalized. Organizations should configure the MITRE ATT&CK connector and build a closed-loop intelligence cycle: OpenCTI TI → MITRE ATT&CK technique mapping → SIEM detection rule → SOC alert. This directly reduces MTTD for campaigns by known threat actors whose TTPs are already documented in the platform.

Recommendation 5 — Establish Analyst Opinion Programs

The Opinion Distribution radar chart visible on all entities is only as valuable as the analyst participation it receives. Organizations should establish a formal process for analysts to submit opinions after investigation — confirming or disputing confidence levels based on first-hand incident data. Over time, this transforms OpenCTI from a passive repository into an actively self-improving intelligence system where field experience directly improves the quality of every data point.

7. Comparative Analysis

7.1 OpenCTI vs. Alternative Platforms

The following table compares OpenCTI against alternative approaches based on evidence gathered during this assessment:

DATA FRESHNESS	Live — March 30, 2026 (same-day IOCs)	2016 BOTS dataset / static logs
TOTAL INDICATORS	686,870 real-time IOCs	~6,044 events (fixed dataset)
AI INTEGRATION	Ariane AI — built-in, context-aware	External ChatGPT (not integrated)
THREAT ACTORS	1,320 documented intrusion sets	Manual mapping only
MALWARE COVERAGE	5,330 entities with variant tracking	Limited / manually maintained
CVE INTEGRATION	2026 CVEs live (CVE-2026-1102 etc.)	No native CVE integration
VISUALIZATION	Knowledge graph + radar charts	Tables and bar charts only
STANDARDS	STIX 2.1 / TAXII 2.1 native	Proprietary formats
REPORT DEPTH	38 APT28 reports; 26 Clop CVEs	N/A
COST	Free / open-source	Commercial license required

OpenCTI's superiority in data freshness, scale, AI integration, and standards compliance makes it the most capable freely accessible threat intelligence platform available as of March 2026. Live 2026 CVE data, Ariane AI, and knowledge graph visualization provide capabilities exceeding many commercial platforms at zero licensing cost.

7.2 Unique Differentiators Observed

- **Same-day IOC ingestion:** Indicators timestamped within hours of the assessment — not common in free platforms
- **Built-in AI (not external):** Ariane answers questions with platform context — no data export to ChatGPT required
- **2026 CVE coverage:** CVE-2026-1102 and CVE-2026-0723 visible — most commercial feeds lag by days to weeks
- **STIX ID visibility:** Every entity shows its machine-readable STIX ID — genuine standards compliance
- **Knowledge graph:** Interactive relationship graph transforms static lists into actionable intelligence maps

8. Conclusion

This assessment of OpenCTI demonstrates that the platform represents the current state-of-the-art in open-source threat intelligence management. Evaluated across seven standardized metrics on March 30, 2026, OpenCTI achieved a Strong rating in every dimension — from Mean Time to Detect through Analyst Feedback and AI Integration.

The platform's live data scale (686,870 indicators, 1,320 threat actors, 5,330 malware entities, 6,390 reports), native STIX 2.1 compliance, built-in Ariane AI, interactive knowledge graphs, and real-time 2026 CVE coverage place it in a category that few commercial platforms can match. The knowledge graph of the APT28 Zebrocy campaign, the Ariane AI response correctly attributing APT28 to GRU Unit 74455, and the 26 Clop CVEs including 2026-dated entries are specific evidence points that demonstrate operational excellence.

The assessment also identified areas for production improvement: connector diversity is essential to avoid single-source dependency, a formal TLP governance policy is required for sensitive intelligence sharing, and analyst training programs must accompany any deployment to realize the platform's full capability.

For organizations building or maturing a Cyber Threat Intelligence program, OpenCTI offers a compelling, cost-effective, technically sophisticated solution that scales from small security teams to enterprise SOCs. Its open-source foundation, combined with Filigran's active commercial roadmap, ensures long-term viability as a cornerstone of any mature threat intelligence operation.

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